

Data Analysis with Python and SQL

Group 23

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How should retail investors invest within the AI vertical?

An analysis of volatility regimes and market indicators

High-Level Takeaways

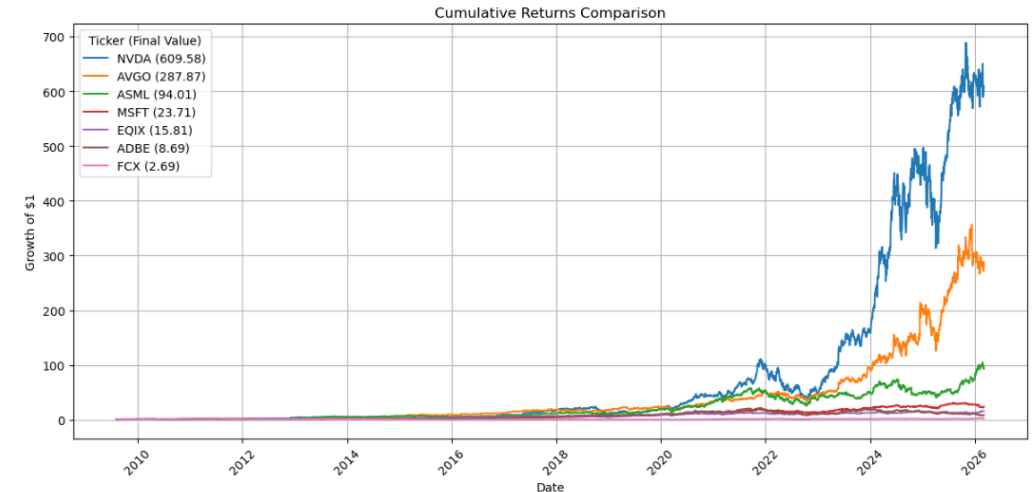
Takeaway 1:

Long-term growth is possible, but a retail investor must be prepared to weather the volatility and drawdowns

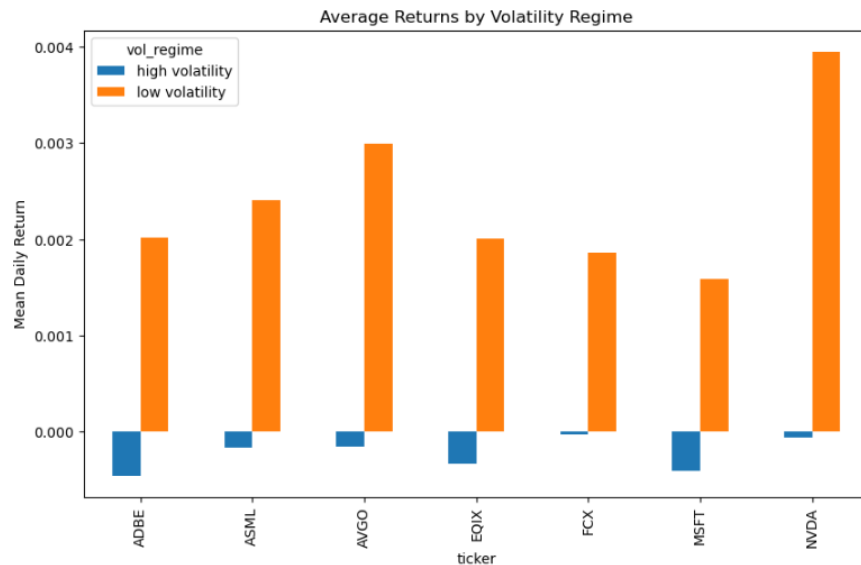
Takeaway 2:

Rolling Return shows the strongest relationship with portfolio return while market volatility has a negative relationship. Overall, the regressions give the expected sentiment that financial returns are difficult to predict.

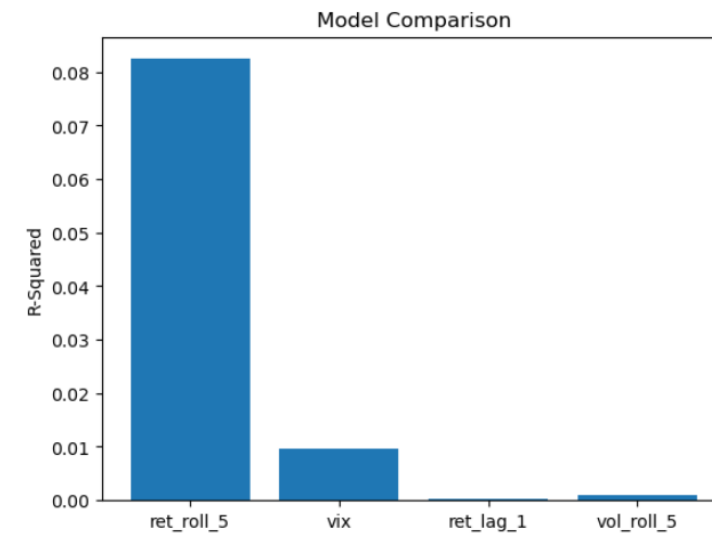
Growth has varied widely from stock to stock suggesting individual companies respond differently to market affects



Market volatility plays a clear role in the overall returns of each stock



Predictability is generally low within the AI vertical



Two forms of market data and a multi-step analysis drove the findings

By joining stock return data with VIX volatility levels and grouping results by ticker, the analysis reveals how performance changes across market regimes.

Data Sources



Stock price dataset (Midterm Data set)



Market Volatility data (VIX Index from FRED)

Data Processing

- Calculate daily stock returns
- Align stock and VIX dates
- Construct equal-weight portfolio returns

Key Insights

Highest Volatility

EQIX

Highest Average Return

NVDA

Largest Downside Risk

EQIX

SQLite Database Structure

- prices → returns → portfolio_returns
- VIX → factors

Analyzing the AI vertical across high and low volatile regimes informs the retail investor what to expect from the portfolio given different market affects

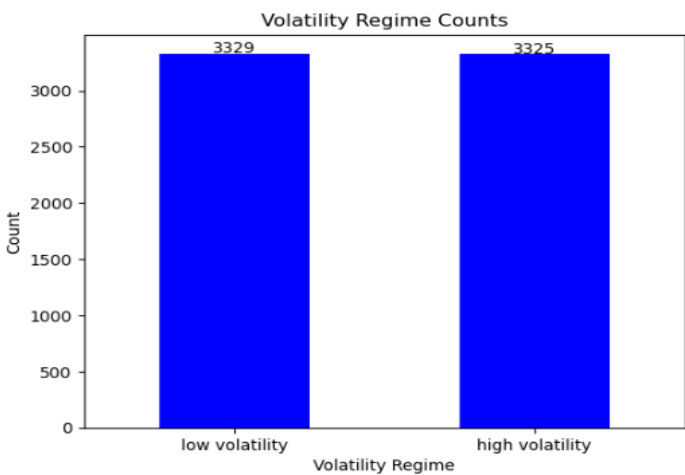
High Volatility

Sector wide forces dominate individual company dynamics when uncertainty rises

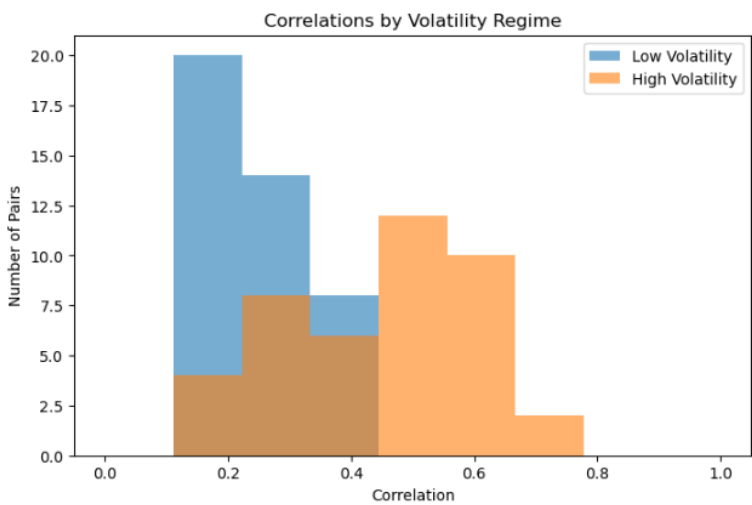
Low Volatility

In contrast, fundamentals tend to differentiate performance and produce positive average returns when volatility is low

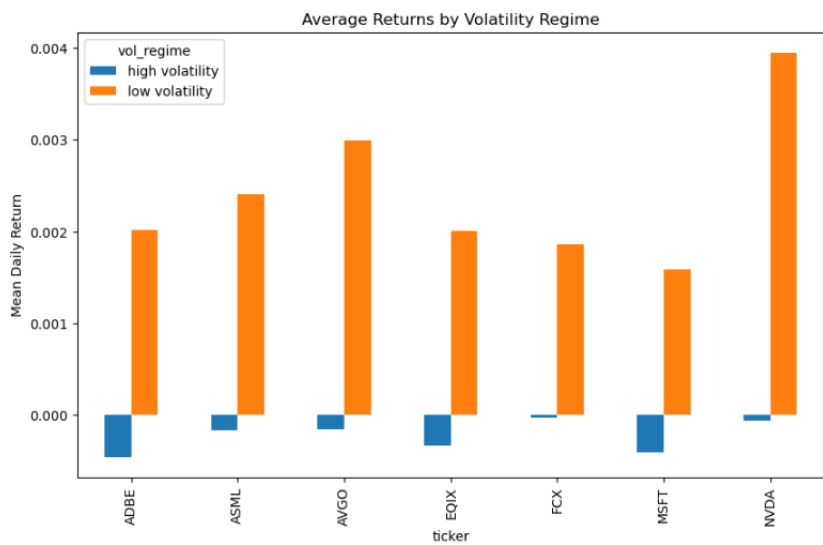
Returns above and below the median produced near identical regimes



The vertical moves together more during high volatility



Low volatility is a driving factor for AI growth

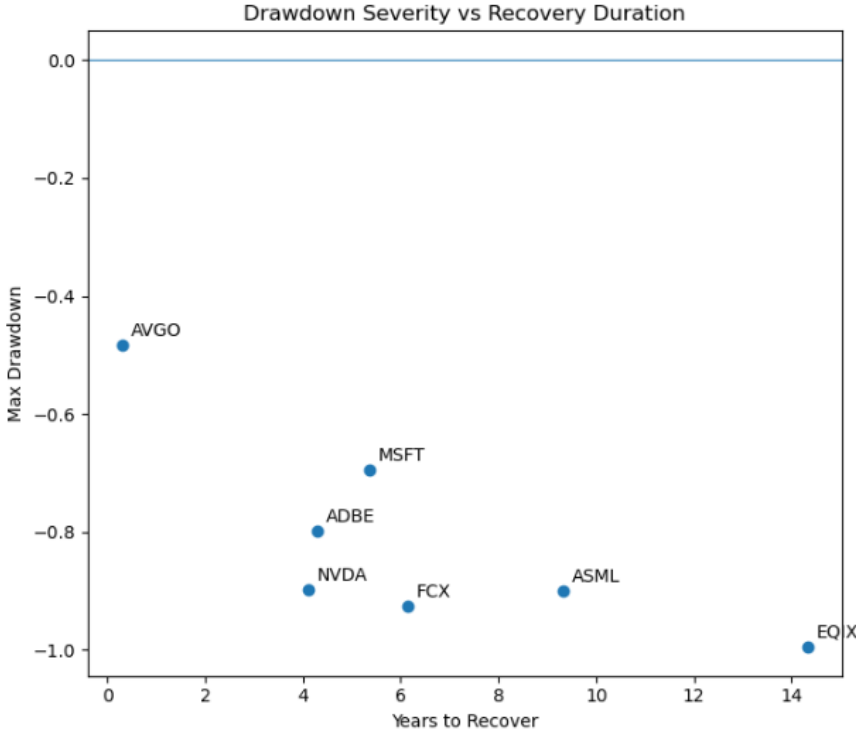


Historical drawdowns require long-term conviction and risk tolerance to realize the benefits of this high growth portfolio

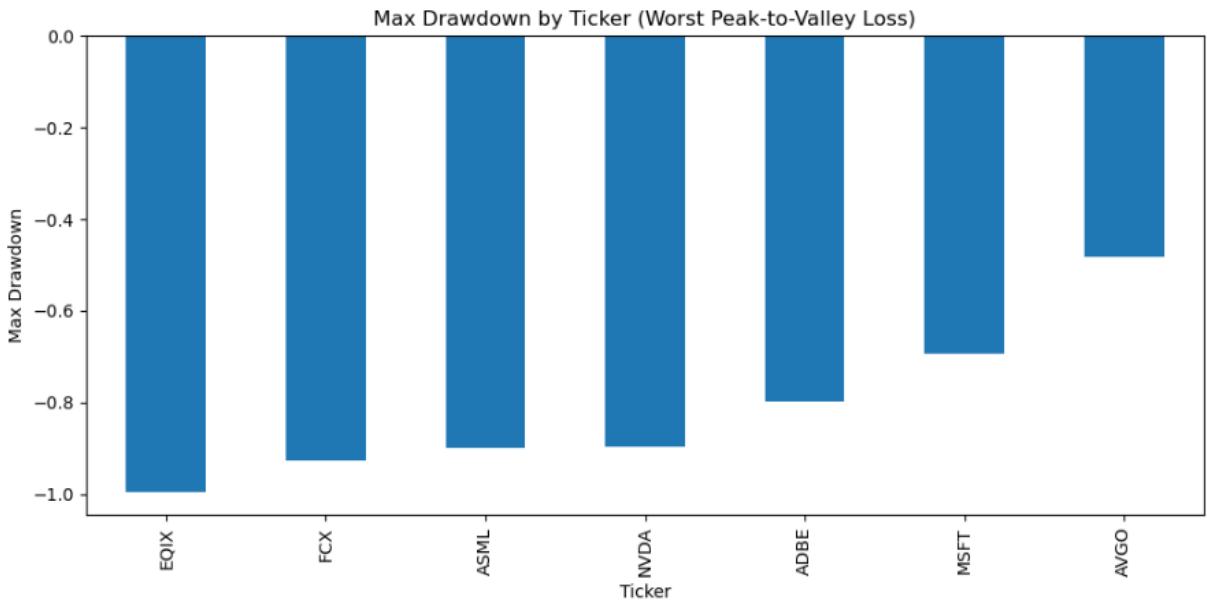
Retail Investor Risk

Retail investors seeking different investing strategies should examine how different portfolio mixes may minimize the effects of steep drawdowns

Nearly all firms took years to recover from their max drawdown

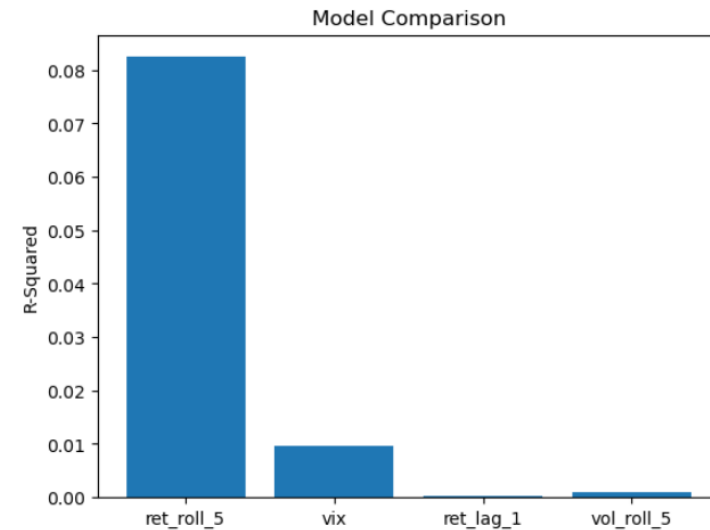
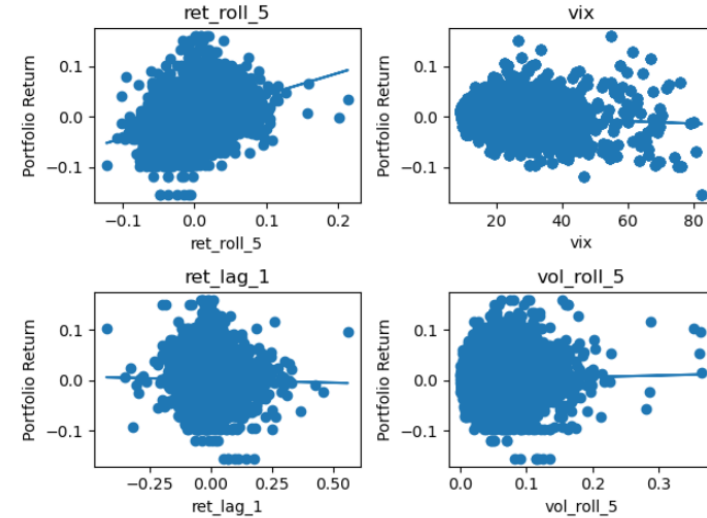


Over 70% of firms lost 80% of their value during this period



Linear Regression Analysis & Setup

- Predictors: Rolling Return, Rolling Volatility, Drawdown, VIX
- Predicting: Portfolio Return
- Highest R-Squared: Rolling Return
- Negative Relationship: VIX, market risk



Logistic Regression Analysis & Setup

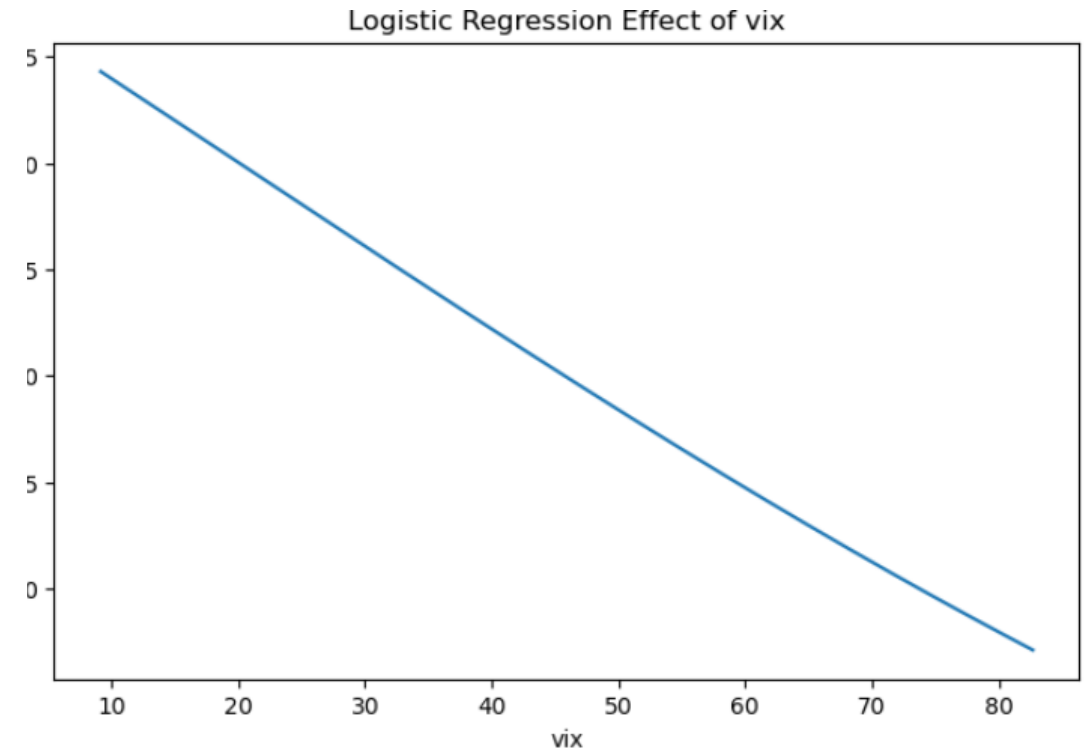
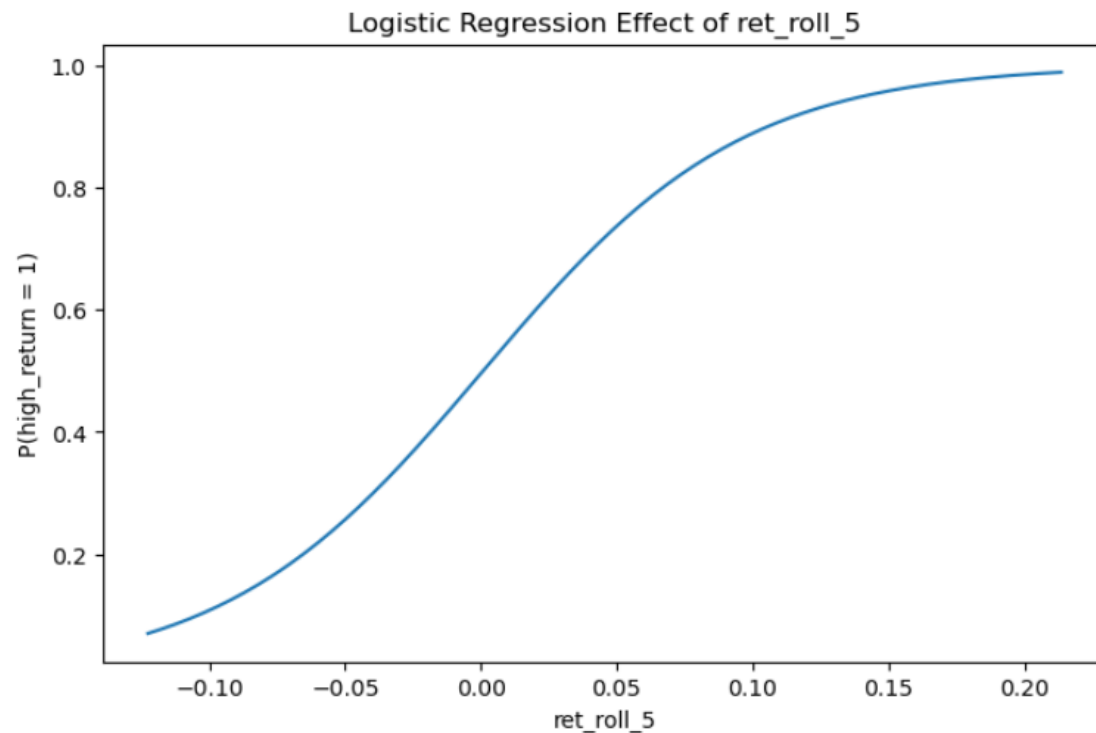
Metrics:

Accuracy: 0.6032

Confusion Matrix:

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[[2441 1991]
 [1476 2830]]
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- Rolling Return + Relationship w/ High Returns
- VIX – Relationship w/ High Returns



Conclusion

- Rolling Return is typically inaccessible to the retail investor.
- Possible Implications:
 - Passive Investing in the AI Vertical
 - Opportunity for gains in sticky returns for the more experienced/dedicated investor.
 - Historical volatility shows need for investor endurance.